

Database Benchmark Report

Automatically generated benchmark report

Date: 2026-07-02 13:57:06

Operating System: Linux

Python: 3.12.3

Databases: H2, HSQLDB

Benchmarks: 8

Measured Runs: 20

Warm-up Runs: 5

Benchmark Summary

Number of databases: **2**

Benchmark categories: **8**

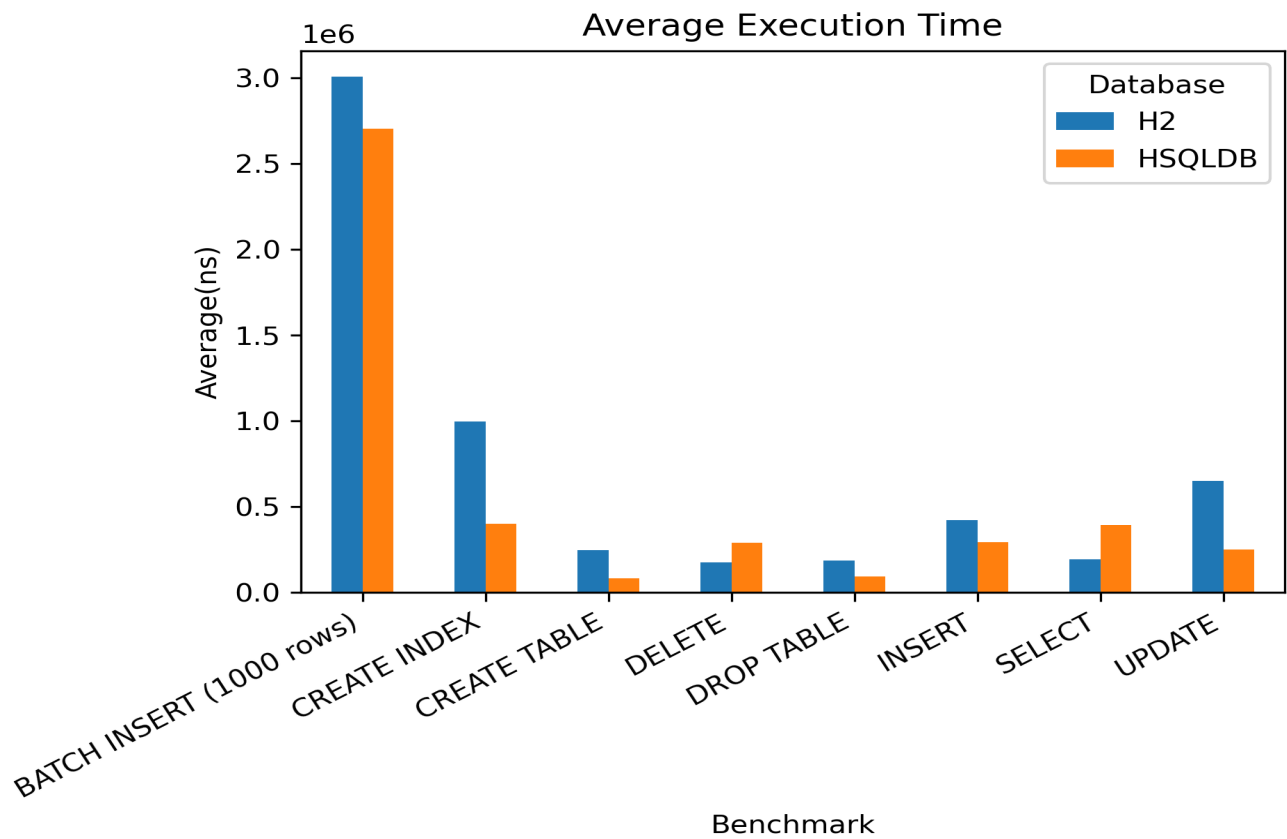
Benchmark results: **16**

This report summarizes execution-time measurements for embedded relational database management systems. All measurements were performed after JVM warm-up iterations and repeated multiple times to improve statistical reliability.

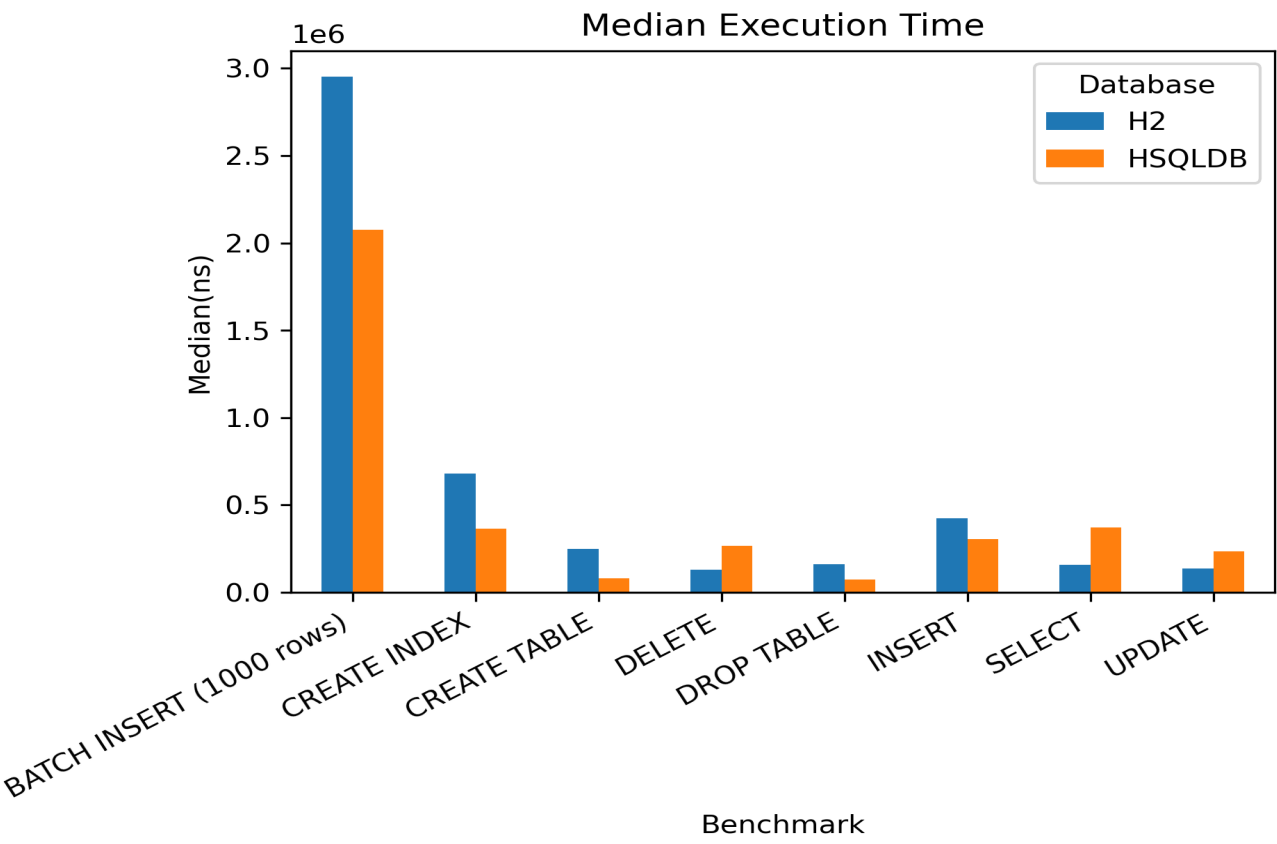
Summary Statistics

Database	Benchmark	Average	Median	Std Dev	Minimum	Maximum
H2	CREATE TABLE	247889	248821	49715	185161	306961
H2	INSERT	420491	423872	178472	174911	658525
H2	BATCH INSERT (1000 rows)	3006769	2951414	246412	2673720	3315193
H2	SELECT	192905	159592	51492	156416	289969
H2	UPDATE	651090	137450	786474	129755	2163353
H2	DELETE	177057	129505	70480	113214	264160
H2	DROP TABLE	184567	162307	57612	117973	255544
H2	CREATE INDEX	996875	680027	445978	649379	1792061
HSQldb	CREATE TABLE	83095	79520	6836	76665	95811
HSQldb	INSERT	294129	304206	46297	236849	351746
HSQldb	BATCH INSERT (1000 rows)	2703860	2075187	1482096	1649301	5643589
HSQldb	SELECT	391703	371603	79600	301841	539561
HSQldb	UPDATE	249911	234785	47895	208404	340024
HSQldb	DELETE	289632	266074	54016	235506	380280
HSQldb	DROP TABLE	92705	75573	39463	66055	171224
HSQldb	CREATE INDEX	402057	365241	76319	312271	531045

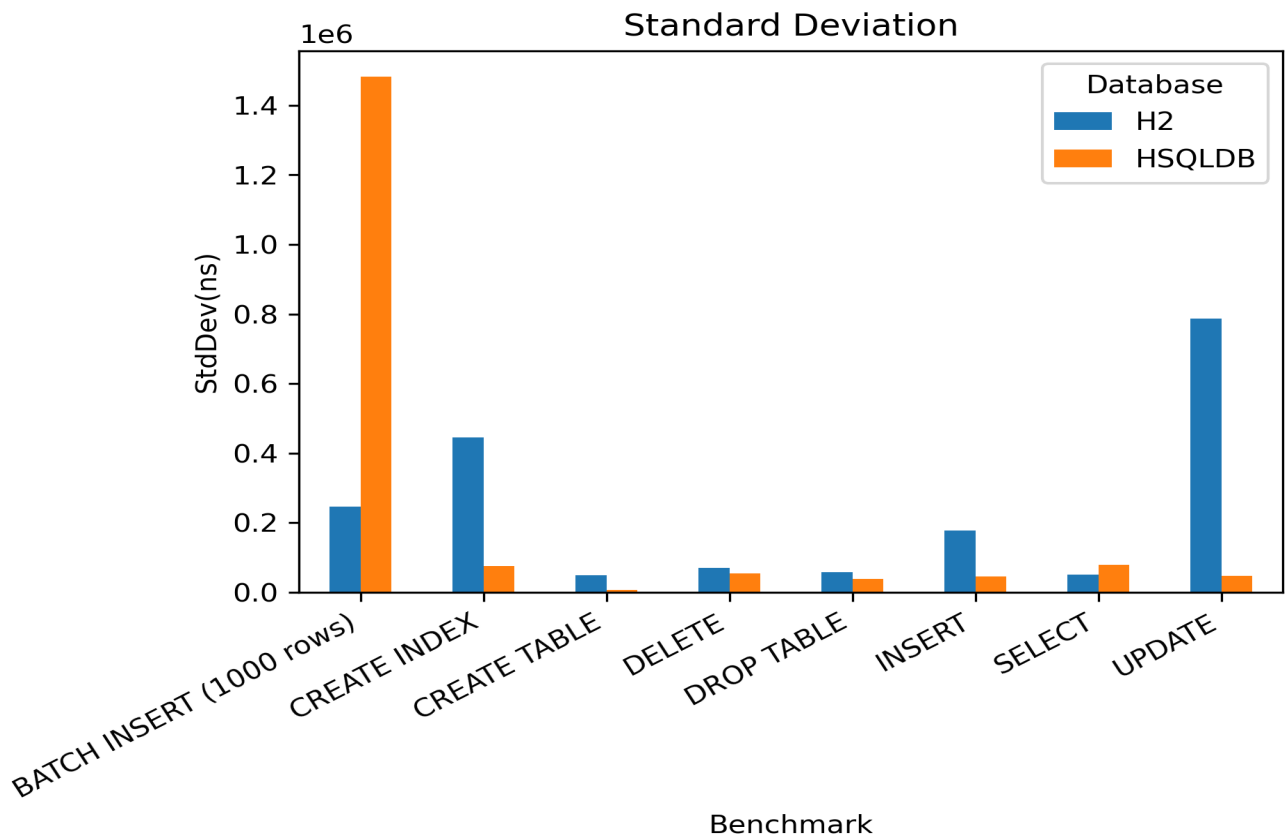
Average Execution Time



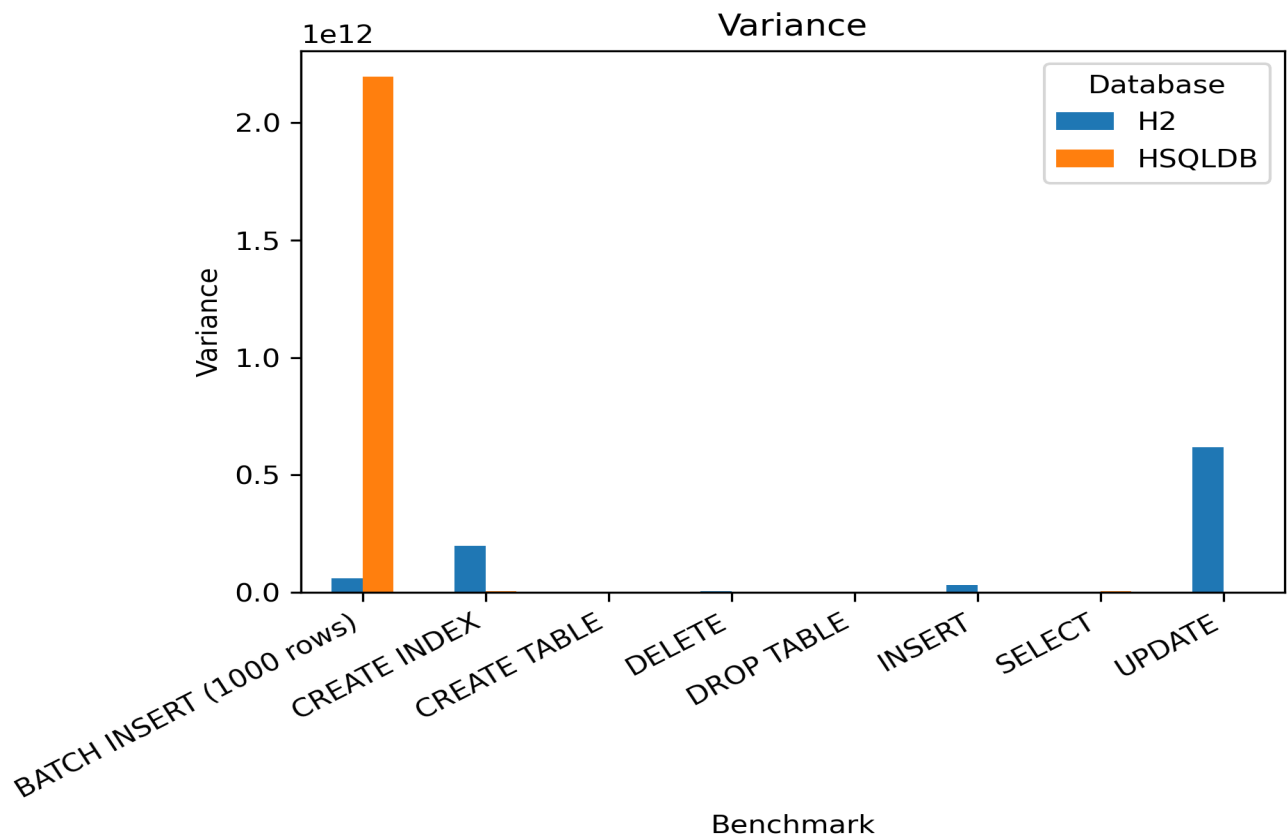
Median Execution Time



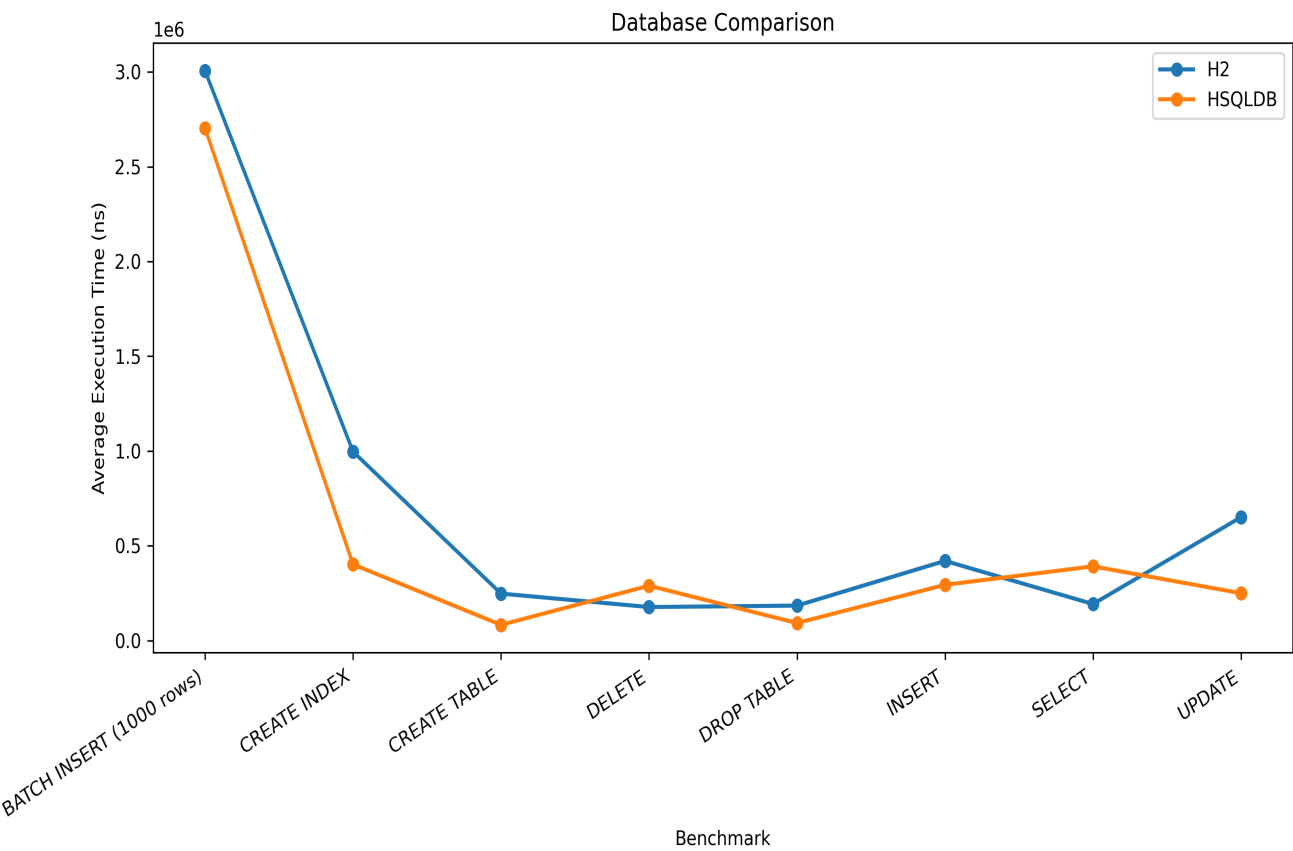
Standard Deviation



Variance



Database Comparison



Benchmark Winners

Benchmark	Winner	Average (ns)
BATCH INSERT (1000 rows)	HSQLDB	2,703,860
CREATE INDEX	HSQLDB	402,057
CREATE TABLE	HSQLDB	83,095
DELETE	H2	177,057
DROP TABLE	HSQLDB	92,705
INSERT	HSQLDB	294,129
SELECT	H2	192,905
UPDATE	HSQLDB	249,911

Overall Benchmark Score

Database	Benchmarks Won
HSQLDB	6
H2	2

Student's t-test

Benchmark	t-statistic	p-value	Significant?
BATCH INSERT (1000 rows)	0.403	0.706380	NO
CREATE INDEX	2.629	0.054966	NO
CREATE TABLE	6.568	0.002439	YES
DELETE	-2.536	0.036816	YES
DROP TABLE	2.631	0.033539	YES
INSERT	1.371	0.234399	NO
SELECT	-4.194	0.004269	YES
UPDATE	1.018	0.365718	NO

Interpretation

A p-value below 0.05 indicates that the observed performance difference between H2 and HSQLDB is statistically significant.

Automatic Conclusion

Summary

The benchmark suite executed 8 benchmark categories on 2 embedded database systems.

Based on the average execution time, **HSQldb** achieved the best overall performance, winning 6 benchmark categories.

The fastest benchmark measured was **CREATE TABLE** executed on **HSQldb** with an average execution time of **83,095 ns**.

The slowest benchmark measured was **BATCH INSERT (1000 rows)** executed on **H2** with an average execution time of **3,006,769 ns**.

Overall, the benchmark results indicate that execution times remained stable after warm-up iterations while demonstrating measurable performance differences between the tested database systems.

This report was generated automatically from the benchmark CSV produced by the Java benchmark framework.